Personal Statement — MSc in Computer Science, The Chinese University of Hong Kong, Shenzhen

The robots I have worked on have often exposed their first serious problems in software rather than in the control law. On the multi-robot platform I develop for at Kaihong, motion-control code could not be reused across hardware until timestamps, coordinate frames and validity checks had been standardised across sensors and actuators; inconsistent state information leads directly to unreliable control. This experience clarified the direction I want to pursue: developing the software, data and distributed-system foundations that allow control and AI algorithms to operate reliably across robot platforms. I have been learning this layer through practical work rather than systematic study, and the MSc in Computer Science is where I intend to build those foundations deliberately.

I am a final-year Computer Science student at Shenzhen Technology University, ranked 6 of 156, with Principles of Automatic Control and Modern Control Theory taken alongside the core curriculum. My computing background has been tested most directly on Kaihong's M-Robots platform on OpenHarmony. I built a unified capability model for sensors, actuators and controllers in C++ and mapped ROS and ROS2 motion commands onto low-level drivers; implemented device discovery, capability registration, state synchronisation and task dispatch over M-DDS; and gated task plans through an action-proposal, policy-check, execution and state-feedback chain. These tasks required interface design, distributed coordination and execution safety, and showed me why platform architecture cannot be learned only through ad hoc project work.

The rest of my work has been control and navigation implemented as software on real hardware. At the Shenzhen Institute of Artificial Intelligence and Robotics for Society I was responsible for the balance control of a dual-arm ballbot, from a deployed PID cascade and LQR analysis of the system as a four-state inverted pendulum to the ROS-to-CAN execution chain and fall-recovery safeguards. In laboratory projects I implemented gain-scheduled LQR and virtual model control on STM32H7/FreeRTOS so that a wheel-legged robot stays balanced as its leg length changes, and the complete EKF/AMCL, Hybrid-A*, TEB and MPC chain that lets a car-like vehicle localise, plan around obstacles and follow its path. In the university RoboMaster team I owned the gimbal, launcher and chassis control loops of the hero robot that reached the 2025 national final. Across these projects, I had to manage real-time constraints, inconsistent timing and coordinate frames in multi-source data, and deliver a complete, documented system rather than an isolated module.

My AI experience is more limited in scale but covers a complete application: an OpenHarmony smart gardening system I built alone, with YOLOv5 inference quantised to INT8 on an RK3588 edge device, a LangGraph agent chain for decisions and a Hi3861 irrigation state machine. It showed me how to place a model within a working product loop; it also showed me that larger fleets require stronger data engineering, evaluation methods and security design than I have formally studied.

The MSc in Computer Science is the right next step because it addresses the computing foundations behind the robot systems I have been building. Its core training in algorithms, artificial intelligence, operating systems and parallel programming, together with electives in machine learning, deep learning, reinforcement learning, computer vision and GPU programming, would help me turn practical robot

experience into a more systematic understanding of computation and intelligent software. I am particularly interested in applying these methods to robot perception, motion planning and data-driven control.

After the degree I intend to work in the Greater Bay Area as a robotics algorithm engineer, combining perception, planning and control with learning-based methods. The programme's project and industry-oriented training would let me study the algorithms behind a complete robot system while testing them under real timing and hardware constraints. My previous work has taken control and navigation modules from software design to working robots; this programme would help me make those algorithms more principled, efficient and transferable across platforms.